

Optional Office Hours Part I Notes

I.0 Errata

I.1 Completeness and Soundness for Sentential Logic

Let Γ and φ be a set of wff's and a wff in sentential logic respectively. To piggyback off of the fact that our first order language only utilizes the connectives $\neg, \rightarrow$, we shall do the same, here, since it shortens a lot of the work, is the perfect bridgeway to the more general first order system, and we can do it without loss of generality since $\{\neg, \rightarrow\}$ is complete as a set of sentential connectives.

Definition I.1.1. We say that φ is **provable** (and write $\Gamma \vdash \varphi$) **through natural deduction from Γ** if there exists a natural deduction using the following inference rules below from Γ to φ :

 $\Gamma \vdash \alpha$ *Id, for any given $\alpha \in \Gamma$*

$\Gamma \vdash \alpha$

 $\Gamma \cup \Gamma' \vdash \alpha$ *W, where Γ' is also a set of wff's*

$\Gamma; \alpha \vdash \beta$

 $\Gamma \vdash \alpha \rightarrow \beta$ *$\rightarrow I$, (analogous to the deduction theorem)*

$\Gamma \vdash \alpha \quad \Gamma \vdash \alpha \rightarrow \beta$

 $\Gamma \vdash \beta$ *$\rightarrow E$, (this one is Modens Ponens!)*

$\Gamma; \varphi \vdash \beta \quad \Gamma; \varphi \vdash \neg\beta$

 $\Gamma \vdash \neg\varphi$ *$\neg I$, (the first reductio ad absurdum rule)*

$\Gamma; \neg\varphi \vdash \beta \quad \Gamma; \neg\varphi \vdash \neg\beta$

 $\Gamma \vdash \varphi$ *$\neg E$, (known as "proof by contradiction")*

Note that proofs in this system are in the form of proof trees that are for example formulated as follows:

Example I.1.2.

To show $(\vee I)$, where $\alpha \vee \beta$ is defined as $\neg\alpha \rightarrow \beta$, we write the proofs as follows:

$$\begin{array}{c}
 \Gamma \vdash \beta \\
 \hline
 \text{-----} W \\
 \Gamma; \neg\alpha \vdash \beta \\
 \hline
 \text{-----} \rightarrow I \\
 \Gamma \vdash \alpha \vee \beta
 \end{array}$$

$$\begin{array}{c}
 \Gamma \vdash \alpha \\
 \hline
 \text{-----} W, \text{-----} Id, \\
 \Gamma; \neg\alpha, \neg\beta \vdash \alpha \quad \Gamma; \neg\alpha, \neg\beta \vdash \neg\alpha \\
 \hline
 \text{-----} \neg E \\
 \Gamma; \neg\alpha \vdash \beta \\
 \hline
 \text{-----} \rightarrow I \\
 \Gamma \vdash \alpha \vee \beta
 \end{array}$$

Theorem I.1.3. (*The Completeness Theorem for Sentential Logic*) $\Gamma \vdash \varphi \iff \Gamma \models \varphi$.

Proving this is divided into a few lemmas:

Lemma I.1.4. (*the Soundness Theorem for Sentential Logic*) $\Gamma \vdash \varphi \implies \Gamma \models \varphi$.

Proof. We proceed by structural induction on the length of the proof tree from Γ to φ . First we have the base case, which has to a proof using only *Id* rule

$$\begin{array}{c}
 \text{-----} Id, \\
 \Gamma \vdash \alpha
 \end{array}$$

since it's the only one where we don't need a prior deduction to infer it. Then we have $\alpha \in \Gamma$, which implies that for all truth assignment mappings v satisfying Γ , we have $\bar{v}(\alpha) = T$, which shows $\Gamma \models \alpha$.

In the inductive step, we have the following cases.

Case W. Suppose that the proof tree ends with the inference rule

$$\frac{\Gamma_0 \vdash \alpha}{\Gamma_0 \cup \Gamma' \vdash \alpha} \text{---} W, \text{ where } \Gamma' \text{ is also a set of wff's}$$

where Γ' is also a set of inference rules and $\Gamma = \Gamma_0 \cup \Gamma'$. By inductive hypothesis, we have $\Gamma_0 \models \alpha$, hence, for any truth assignment mapping v satisfying Γ , the truth assignment mapping satisfies the subset Γ_0 , hence satisfies α , which completes this case.

Case $\rightarrow I$. Suppose that the proof tree ends with the inference rule

$$\frac{\Gamma; \alpha \vdash \beta}{\Gamma \vdash \alpha \rightarrow \beta} \rightarrow I,$$

Then by inductive hypothesis we have $\Gamma; \alpha \models \beta$. Note that we proved in previous exercises that this implies $\Gamma \models \alpha \rightarrow \beta$, and this case is complete.

Case $\rightarrow E$. Left to the reader.

Case $\neg I$. Suppose that the proof tree ends with the inference rule

$$\frac{\Gamma; \varphi \vdash \beta \quad \Gamma; \varphi \vdash \neg \beta}{\Gamma \vdash \neg \varphi} \neg I,$$

We find by inductive hypothesis that $\Gamma; \varphi \models \beta, \Gamma; \varphi \models \neg \beta$, which shows that $\Gamma; \varphi$ is unsatisfiable. We find by a Corollary in Enderton that $\Gamma; \varphi$ being unsatisfiable implies that $\Gamma \models \neg \varphi$ (and intuitively this holds since we cannot formulate a variable assignment function that yields truth in both β and $\neg \beta$).

%MORE DIRECTLY CITE SOURCES HERE

Case $\neg E$. Left to the reader. $\square$

We can define the consistency of Γ analogously to the definition of consistency in the more general first order case

Definition 1.1.6.

1. We say that Γ is **consistent** if $\Gamma \not\vdash \neg \beta \wedge \beta$, for some wff β

2. We say that Γ set **maximally consistent** if Γ is consistent and either $\varphi \in \Gamma$ or $\neg\varphi \in \Gamma$, but not both.

Lemma I.1.5. (*Lindenbaun Lemma*) For every consistent Γ , there exists a maximally consistent Γ' , i.e., a consistent Γ' such that for every wff φ , we have either $\varphi \in \Gamma'$ or $\neg\varphi \in \Gamma'$ (but never both).

Outline of the proof. This essentially follows by a similar construction of Δ_n as done in the proof of **Lemma 6.2.4** of the notes, except that we add σ_n if $\Delta_n \cup \{\sigma_n\}$ is still consistent, and we don't otherwise. Details will be left to the reader. $\square$

Lemma I.1.6. If Γ is consistent, then Γ is satisfiable, i.e., there exists a truth assignment mapping v such that $\bar{v}(\phi) = T$, for all $\phi \in \Gamma$.

Proof. Note by the previous lemma that there exists a maximally consistent set Γ' . Next we proceed as we did with **Lemma 6.2.5** of the notes, and define $v_{\Gamma'}$ by

$$v_{\Gamma'}(\sigma) = T \iff \sigma \in \Gamma'.$$

We find that $v_{\Gamma'}$ is a well-defined truth assignment mapping satisfying Γ , details of which shall be left to the reader. $\square$

Proof of Theorem I.1.3. Note that $\implies$ is simply just soundness proved in **Lemma I.1.4**.

For $\impliedby$, it shall suffice by **Lemma I.1.6** to prove that the statement

Γ is consistent $\implies \Gamma$ is satisfiable (given in the previous lemma)

is equivalent to the statement

$$\Gamma \models \varphi \implies \Gamma \vdash \varphi$$

To prove $\implies$, note that by hypothesis, we have contrapositively that Γ is unsatisfiable $\implies \Gamma$ is inconsistent. It immediately follows that

$$\begin{aligned} \Gamma \models \varphi &\implies \Gamma; \neg\varphi \text{ is unsatisfiable} \\ &\implies \Gamma; \neg\varphi \text{ is inconsistent} \\ &\implies \Gamma; \neg\varphi \vdash \beta \text{ and } \Gamma; \neg\varphi \vdash \neg\beta, \text{ for some wff } \beta \end{aligned}$$

$\implies \Gamma \vdash \varphi$ using $\neg E$ inference rule.

To prove $\Leftarrow$, note that the contrapositive Γ is unsatisfiable $\implies \Gamma$ is inconsistent is a special case of $\Gamma \models \varphi \implies \Gamma \vdash \varphi$, since it boils down to proving that $\Gamma \models \beta$ and $\Gamma \models \neg\beta$ for some wff $\beta \implies \Gamma \vdash \beta$ and $\Gamma \vdash \neg\beta$. $\square$

I.2 Effectiveness/Computability

%EDIT THIS SECTION TO MAKE IT MORE CLEAR BEFORE GODEL'S INCOMPLETENESS THEOREM

Definition I.2.1. We define an "**effective procedure**" to be any algorithm that we can execute in a computer program.

that finishes after a finite amount of time, i.e., the program halts.

%BE MORE SPECIFIC ABOUT HALTING

Definition I.2.2. For any set A (usually a subset of $\mathbb{N}^k$), we call A **decidable** if there exists an effective procedure that determines whether or not $\alpha \in A$.

Example I.2.3.

(i) Note that the whole set $\mathbb{N}$ and the $\emptyset$ are decidable since we can input any $n \in \mathbb{N}$ and automatically conclude that $n \in \mathbb{N}$ or $n \notin \mathbb{N}$, respectively.

(ii) The so-called "golden ratio set" A recursively defined as the sequence $A := \{a_0, a_1, a_2, \dots\}$ recursively defined by $a_0 := 0, a_1 := 1, a_{n+2} := a_n + a_{n+1}$, for all $n \geq 0$. This is decidable since given $N \in \mathbb{N}$, we can check to see if $N = a_n$, for some $n \in \mathbb{N}$, by going through each $a_n \leq N$ and checking whether $a_n = N$ until we reach $a_n > N$ (in which case, if we don't find anything, we conclude that $N \notin A$).

Definition I.2.4. For any set A (usually again a subset of $\mathbb{N}^k$), we call A **semidecidable** or **recursively enumerable** if there exists an effective procedure that for each $\alpha \in A$, shows that α in fact is in A . However this procedure may be indeterminate (and particularly be an algorithm that doesn't terminate) if $\alpha \notin A$.

Example I.2.5.

(i) Any decidable set is by definition semidecidable by definition, so the previous examples

mentioned are also semidecidable.

(iii) Let A be the set of programs that halt when executed. We find A is semidecidable, since given $\alpha \in A$, we can simply execute the program using a compiler (as we usually do in an introduction to computer programming class), and if it halts, then we execute the program and it halts which shows that α is in A . However, if $\alpha \notin A$, then α will not stop. The conjecture on whether or not this set is decidable is called the Halting Problem, and it was proved by Alan Turing that the set A is in fact not decidable, giving us a counterexample of something that is semidecidable but not decidable.

%CITE LINK TO HALTING PROBLEM

The following partial converse to decidability from semidecidability, however, does hold:

Theorem 1.2.6. If $A \subset \mathbb{N}^k$, and A and $\mathbb{N}^k \setminus A$ are semidecidable, then A (and $\mathbb{N}^k \setminus A$) are decidable.

Outline of proof. Choose two programs p_1, p_2 that halt when $\alpha \in A$ and $\alpha \notin A$, respectively. Let p be the program that alternates between running each step of p_1 and p_2 , and note that if $\alpha \in A$, then p halts since p_1 halts, and if $\alpha \notin A$, p halts since p_2 halts. Either way, p is an effective procedure determining whether or not $\alpha \in A$. $\square$

Next time in Optional Office Hours: We'll talk about how decidability applies to interesting questions we'd like to ask about the proof/semantic system in both sentential logic and first order logic, so STAY TUNED!!!